Strengths:
- Vast model repository with thousands of pretrained models.
- Simple APIs for inference and training.
- Strong community support and extensive documentation.
- Supports multiple frameworks, including PyTorch and TensorFlow.

Use cases:
- Text generation and summarisation
- Sentiment analysis
- Question answering and conversational AI

Hugging Face hosts a vast collection of pretrained models, such as BERT, GPT-2, T5, and BLOOM. Users can fine-tune these models on custom datasets using transfer learning, reducing computational costs compared to training from scratch.

OpenAI's GPT models (via open source APIs)
Although OpenAI's latest GPT models are not fully open source, it provides APIs for accessing models like GPT-4 and GPT-3.5. Community-driven implementations, such as GPT-J by EleutherAI, attempt to provide open alternatives.

Key points:
- OpenAI's API provides access to powerful models but requires usage-based licensing.
- Open implementations like GPT-NeoX offer community-driven alternatives.
- GPT models excel in text completion, summarisation, and conversational AI applications.

PyTorch and TensorFlow
PyTorch and TensorFlow are foundational deep-learning frameworks used for developing LLMs from scratch or fine-tuning existing ones. Table 2 compares them.

Feature	PyTorch	TensorFlow
Flexibility	Dynamic computation graph for flexibility	Optimised for large scale production deployments
Adoption	Popular among researchers and academics	Backed by Google with strong enterprise adoption
Ecosystem	Strong Hugging Face integration	Extensive tools for model optimisation and serving

Table 2: A comparison of PyTorch and TensorFlow

LangChain
LangChain is an open source framework designed to build LLM-powered applications by integrating different components like memory, tools, and agents.

Features:
- Simplifies building AI-powered workflows.
- Provides modules for context-aware reasoning and tool use.
- Compatible with multiple LLM providers, including OpenAI and Hugging Face.

Use cases:
- Chatbots with memory
- Document summarisation
- Autonomous AI agents

Rasa
Rasa is an open source framework tailored for building AI-driven chatbots and virtual assistants.

Key features:
- NLU (natural language understanding) and dialogue management.
- On-premises deployment for data privacy.
- Customisable and extensible architecture.

Use cases:
- Customer service automation
- Enterprise chatbots
- Voice assistants

Table 3 lists several other projects that contribute to the open source LLM landscape.

EleutherAI (GPT-Neo, GPT-J, GPT-NeoX)	Open source alternatives to OpenAI's GPT models.
Cohere	Provides API-based access to large language models.
BLOOM	A multilingual open source LLM developed by the BigScience initiative.

Table 3: Other open source LLM frameworks

Training and fine-tuning large language models

Fine-tuning basics
Fine-tuning an LLM involves training a pre-trained model on a smaller, task-specific dataset to improve performance on a particular use case. This is useful when:
- The base model is too general and lacks domain-specific knowledge.
- Performance on a specific task is suboptimal using a pre-